文杰 Homework 3 123090612

Q1

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This question is a bit text heavy, it's easier for me to write it in LaTeX..
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a) False.

In the two-phase simplex method, phase I constructs an auxiliary problem to find a feasible solution to the original LP. The auxiliary problem is typically:

$$\min \sum_{i=1}^{m} y_i \quad \text{subject to} \quad Ax + y = b, \quad x \geq 0, \quad y \geq 0,$$

where $y \in \mathbb{R}^m$ are artificial variables. This problem has $n + m$ variables (n original variables x and m artificial variables y) and m equality constraints. The simplex method, used in phase I, seeks a basic feasible solution (BFS) that minimizes the sum of the artificial variables.

A BFS for this auxiliary problem is defined by selecting m linearly independent columns from the constraint matrix $[A, I]$ (where I is the $m \times m$ identity matrix), setting the remaining $n + m - m = n$ variables (non-basic) to zero, and solving for the m basic variables. Thus, a BFS has exactly m basic variables, which may be non-zero, and at most m non-zero entries, because non-basic variables are zero. In degenerate cases, some basic variables may be zero, reducing the number of non-zero entries below m, but it cannot exceed m.

Since the simplex method in phase I terminates with a BFS of the auxiliary problem—whether the original problem is feasible ($\sum y_i^* = 0$) or infeasible ($\sum y_i^* > 0$)—the solution (x^*, y^*) is a BFS. Therefore, (x^*, y^*) has at most m non-zero entries and cannot have more than m.

While the optimal solution set may include such points in theory, the simplex method specifically delivers a BFS. Thus, for the solution (x^*, y^*) obtained in phase I, the number of non-zero entries is at most m.

b) False.

Phase I solves the auxiliary problem (shown in `(a)`) to determine the feasibility of the original problem's feasible region $\{x : Ax = b, x \geq 0\}$ ——

- If the optimal value is zero ($y^* = 0$), the original problem is feasible, and x^* is a feasible solution.
- If the optimal value is positive, the original problem is infeasible.

Unboundedness in an LP means the feasible region is non-empty, and the objective $c^T x$ can be made arbitrarily small. However, phase I focuses solely on feasibility, not the behavior of the original objective function $c^T x$. The auxiliary problem's objective, $\sum y_i$, is bounded below by zero since $y_i \geq 0$. Thus, the auxiliary problem cannot be unbounded below; the simplex method will either find a minimum (zero or positive) or indicate that the auxiliary problem has no feasible solution (though this is rare, as an initial BFS often exists, e.g., $x = 0, y = b$ if $b \geq 0$, with adjustments for $b_i < 0$).

Since phase I only concludes feasibility or infeasibility, it does not assess whether $c^T x$ is unbounded over the feasible region. Thus, phase I cannot indicate unboundedness of problem (1).

c) False.

A basic feasible solution (BFS) to $Ax = b, x \geq 0$ has m basic variables (corresponding to m linearly independent columns of A) and $n - m$ non-basic variables set to zero. In a non-degenerate BFS, all m basic variables are positive; in a degenerate BFS, some may be zero, reducing the number of positive components below m. The statement assumes x^* has fewer than m positive components and asks if there exists another optimal BFS with exactly m positive entries (implying non-degeneracy).

In LP, if an optimal solution exists, at least one is a BFS (a vertex of the feasible region). However, whether an optimal BFS with exactly m positive entries exists depends on the problem.

Consider this counterexample:

- **Constraints:** $x_1 + x_2 = 1$, $x_1 + x_3 = 1$, $x_1, x_2, x_3 \geq 0$
- **Matrix:** $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, so $m = 2$, $n = 3$
- **Objective:** Minimize $-x_1$
  Feasible region: $x_2 = 1 - x_1$, $x_3 = 1 - x_1$, $x_1 \in [0, 1]$. The objective $-x_1$ is minimized (i.e., x_1 maximized) at $x^* = (1, 0, 0)$, with value -1. This has 1 positive component, less than $m = 2$.

BFSs:

1. $x_1 = 1, x_2 = 0, x_3 = 0$ (basis: columns 1, 2 or 1, 3; degenerate, as $x_2 = 0$ or $x_3 = 0$), objective = -1
2. $x_1 = 0, x_2 = 1, x_3 = 1$ (basis: columns 2, 3), objective = 0

Only $x^* = (1, 0, 0)$ is optimal, with 1 positive entry. The BFS $(0, 1, 1)$ has 2 positive entries but is not optimal ($0 > -1$). Since the optimal value is uniquely -1, no other optimal solution exists, including a BFS with exactly $m = 2$ positive entries.

∴ The statement fails when the optimal solution is unique and degenerate or when no non-degenerate optimal BFS exists.

d) True.

A BS arises by selecting m linearly independent columns of A (a basis A_B), setting the $n - m$ non-basic variables to zero, and solving $A_B x_B = b$. If $x_B \geq 0$, it's a BFS and an extreme point of the feasible region. If any x_B component is negative, it's infeasible.

Extreme points are precisely the BFSs—feasible BSs. Thus, a BS that is not an extreme point must have some basic variables negative, making it infeasible.

∴ The statement is **true**, as basic solutions not at extreme points are infeasible.

#2 After introducing slack variables:

$$\begin{aligned}
\text{maximize} \quad & X_1 + 2X_2 + 3X_3 + 4X_4 + 0S_1 + 0S_2 + 0S_3 \\
\text{subject to:} \quad & X_1 + X_2 + X_3 + S_1 = 1 \\
& X_3 - X_4 + S_2 = 2 \\
& X_2 - 2X_3 + 4X_4 + S_3 = 3 \\
& X_1, X_2, X_3, X_4, S_1, S_2, S_3 \geq 0
\end{aligned}$$

For basis $\{5,6,7\}$, with BFS $x = [0\ 0\ 0\ 0\ 1\ 2\ 3]^T$

B	-1	-2	-3	-4	0	0	0
5	1	1	1	0	1	0	0	1
6	0	0	1	-1	0	1	0	2
7	0	1	-2	4	0	0	1	3

⇒

New basis $\{1,6,7\}$, with BFS $x = [1\ 0\ 0\ 0\ 0\ 2\ 3]^T$

B	0	-1	-2	-4	0	0	0	1
1	1	1	1	0	1	0	0	1
6	0	0	1	-1	0	1	0	2
7	0	1	-2	4	0	0	1	3

⇒

New basis $\{2,6,7\}$, with BFS $x = [0\ 1\ 0\ 0\ 0\ 2\ 3]^T$

B	1	0	-1	-4	2	0	0	2
2	1	1	1	0	1	0	0	1
6	0	0	1	-1	0	1	0	2
7	-1	0	-3	4	-1	0	1	2

New basis $\{3,6,7\}$, with BFS $x = [0\ 0\ 1\ 0\ 0\ 1\ 5]^T$

B	2	1	0	-4	3	0	0	3
3	1	1	1	0	1	0	0	1
6	-1	-1	0	-1	-1	1	0	1
4	2	3	0	4	2	0	1	5

⇒

New basis $\{3,4,6\}$, with BFS $x = [0\ 0\ 1\ \frac{5}{4}\ 0\ \frac{9}{4}\ 0]^T$

B	4	4	0	0	5	0	0	8
3	1	1	1	0	1	0	0	1
6	$-\frac{1}{2}$	$-\frac{1}{4}$	0	0	$-\frac{1}{2}$	1	$\frac{1}{4}$	$\frac{9}{4}$
4	2	3	0	4	2	0	1	5

⇒

∴ $x = [0\ 0\ 1\ \frac{5}{4}\ 0\ \frac{9}{4}\ 0]^T$, the optimal value is 8.

#3 ∵ $X_1 - X_2 \geq 1$, $X_1 \geq 1 + X_2$, $X_2 \geq 0$ ∴ $X_1 \geq 0$

∵ $X_1 + X_2 - X_3 - X_4 \leq 2$ ∴ $X_4 \geq X_1 + X_2 - X_3 - 2$

After transfer into standard form:

$$\begin{aligned}
\text{minimize} \quad & X_1 + X_2 + 2X_4^+ - 2X_4^- \\
\text{subject to:} \quad & X_1 - X_2 - S_1 = 1 \\
& X_1 + X_2 - X_3 - X_4^+ + X_4^- + S_2 = 0 \\
& X_1, X_2, X_3, X_4^+, X_4^-, S_1, S_2 \geq 0
\end{aligned}$$

phase I:

$$\begin{aligned}
\text{minimize} \quad & S_3 + S_4 \\
\text{subject to:} \quad & X_1 - X_2 - S_1 + S_3 = 1 \\
& X_1 + X_2 - X_3 - X_4^+ + X_4^- + S_2 + S_4 = 0 \\
& X_1, X_2, X_3, X_4^+, X_4^-, S_1, S_2, S_3, S_4 \geq 0
\end{aligned}$$

For basis $\{7,8\}$, with BFS $x=[0\ 0\ 0\ 0\ 0\ 0\ 1\ 2]^T$

B	-2	0	1	1	1	-1	0	0	-3
7	1	-1	0	0	-1	0	1	0	1
8	1	1	-1	-1	0	1	0	1	2

⇒

New basis $\{1,8\}$ with BFS $x=[1\ 0\ 0\ 0\ 0\ 0\ 0\ 1]^T$

B	-2	0	1	1	1	-1	0	0	-1
1	1	-1	0	0	-1	0	1	0	1
8	0	2	-1	-1	1	1	-1	1	1

⇒

New basis $\{1,2\}$ with BFS $x=[\frac{3}{2}\ \frac{1}{2}\ 0\ 0\ 0\ 0\ 0]^T$

B	0	0	0	0	0	0	1	1	0
1	1	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{3}{2}$
2	0	1	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

∴ The original problem is feasible. $x=[\frac{3}{2}\ \frac{1}{2}\ 0\ 0\ 0\ 0\ 0]^T$

phase II: $C^T=[1\ 1\ 0\ 2\ 0\ 0]$, $A=\begin{bmatrix}1&-1&0&0&-1&0\\1&1&-1&-1&0&1\end{bmatrix}$, $b=\begin{bmatrix}1\\2\end{bmatrix}$

$\bar{C}^T=C^T-C_B^TA_B^{-1}A=[0\ 1\ \frac{1}{2}\ \frac{5}{2}\ \frac{1}{2}\ -\frac{1}{2}]$

New basis $\{1,6\}$, with BFS $x=[1\ 0\ 0\ 0\ 0\ 1]^T$

B	0	2	0	2	1	0	-1
1	1	-1	0	0	-1	0	1
2	0	2	-1	-1	1	1	1

∴ The optimal value is 1, $x=[1\ 0\ 0\ 0\ 0\ 1]^T$

#4 After transfer into standard form:

minimize $x_2-4x_3+x_5$
subject to: $x_1+x_2+x_3+x_4=1$
$x_2+x_3+x_4-x_5=-1$
$x_1+2x_3-x_5-s_1=0$
$x_1,x_2,x_3,x_4,x_5,s_1\geq 0$

phase I:

minimize $s_2+s_3+s_4$
subject to: $x_1+x_2+x_3+x_4+s_2=1$
$-x_2-x_3-x_4+x_5+s_3=1$
$x_1+2x_3-x_5-s_1+s_4=0$
$x_1,x_2,x_3,x_4,x_5,s_1,s_2,s_3,s_4\geq 0$

For basis $\{7,8,9\}$, with BFS $x=[0\ 0\ 0\ 0\ 0\ 0\ 1\ 1\ 0]^T$

B	-2	0	-2	0	0	1	0	0	0	-2
7	1	1	1	1	0	0	1	0	0	1
8	0	-1	-1	-1	1	0	0	1	0	1
9	1	0	2	0	-1	-1	0	0	1	0

⇒

New basis $\{1,8,9\}$ with BFS $x=[1\ 0\ 0\ 0\ 0\ 0\ 0\ 1\ -1]^T$
⇒ infeasible

New basis $\{1,5,6\}$ with BFS $x=[1\ 0\ 0\ 0\ 1\ 1\ 0\ 0\ 0]^T$

$s_2+s_3+s_4=0$ ∴ $s_2,s_3,s_4=0$

∴ The original problem is feasible.

phase II: For basis $\{1,5,6\}$, $C^T=[0\ 1\ -4\ 0\ 1\ 0]$, $A=\begin{bmatrix}1&1&1&1&0&0\\0&-1&-1&-1&1&0\\1&0&2&0&-1&-1\end{bmatrix}$, $b=\begin{bmatrix}1\\1\\0\end{bmatrix}$

$\bar{C}^T=[0\ 2\ -3\ 1\ 0\ 0]$

New basis $\{3,5,6\}$ with BFS $x=[0\ 0\ 1\ 0\ 2\ 0]^T$

B	3	5	0	4	0	0	2
3	1	1	1	1	0	0	1
5	1	0	0	0	1	0	2
6	1	2	0	2	1	1	2

∴ The optimal value is -2, $x=[0\ 0\ 1\ 0\ 2\ 0]^T$

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Q5

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This question is a bit text heavy, it's easier for me to write it in LaTeX..
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a) The Current Solution is Feasible but Not Optimal

Feasibility: The current solution is feasible if all basic variables are non-negative:

- $x_1 = 0 \geq 0$,
- $x_4 = \eta \geq 0$, so $\eta \geq 0$,
- $x_5 = 5 > 0$.
  $\therefore \eta \geq 0$ ensures feasibility.

Not Optimal: The solution is not optimal if the objective can be improved. For minimization, this occurs if any non-basic variable has a negative reduced cost, allowing z to decrease. Here:

- x_7 has reduced cost -1 < 0, regardless of parameter values.

To confirm improvement is possible, check x_7 's column in the constraint rows: [0, δ, 0].

- If $\delta > 0$, increasing x_7 affects $x_5 = 5 - \delta x_7$, and x_7 can increase to $5/\delta$, making $z = -5/\delta < 0$.
- If $\delta \leq 0$, $x_5 = 5 - \delta x_7$ remains non-negative as x_7 increases indefinitely, making $z \to -\infty$.
  $\Rightarrow$ In both cases, the solution is not optimal.
  $\therefore \eta \geq 0$ is sufficient, as the negative reduced cost of x_7 ensures non-optimality.

Condition: $\eta \geq 0$.

b) The LP is Unbounded (i.e., the Optimal Value is $-\infty$)

The LP is unbounded if z can decrease to $-\infty$ from a feasible solution. This occurs when a non-basic variable with negative reduced cost can increase indefinitely without making any basic variable negative, i.e., all coefficients in its constraint column are ≤ 0.
$\therefore$ When $A_{ij} \leq 0$ to every i -th row in the incoming colunm, the problem is unbounded.

1. When col `3` is the incoming column: α<0, $\gamma \leq 0$, the problem is unbounded.
2. When col `6` is the incoming column: $\alpha \geq 0, \beta$<0, η<0.
3. When col `7` is the incoming column: $\alpha \geq 0, \beta \geq 0, \delta$<0.

c) The Objective Function at the New Basic Feasible Solution After One Simplex Iteration is -1

Perform one iteration by choosing a non-basic variable with negative reduced cost:

- **Enter** x_7: Reduced cost = -1. Column: [0, δ, 0]. If $\delta > 0$, pivot on δ (row 5, x_5):
  - $x_5 = 5 - \delta x_7$, so $x_7 = 5/\delta$, $x_5 = 0$.
  - $x_4 = \eta$, $x_1 = 0$.
  - New basic variables: $x_1 = 0$, $x_4 = \eta$, $x_7 = 5/\delta$.
  - $z = -1 \cdot x_7 = -5/\delta$. Set $-5/\delta = -1$, so $\delta = 5$.
- **Enter** x_6 (if $\beta < 0$): Column: [1, 0, -1]. Pivot on 1 (row 4, x_4, if $\eta > 0$):
  - $x_4 = \eta - x_6$, so $x_6 = \eta$, $x_4 = 0$.
  - $x_1 = x_6 = \eta$, $x_5 = 5$.
  - New basic variables: $x_1 = \eta$, $x_5 = 5$, $x_6 = \eta$.
  - $z = \beta \cdot x_6 = \beta\eta$. Set $\beta\eta = -1$, e.g., $\eta > 0$, $\beta = -1/\eta$.

d) The Simplex Method Stops in the Next Iteration

1. If $\alpha \geq 0$ and $\beta \geq 0$ and $\delta \geq 0$: $\alpha - \frac{1}{\delta}$>0, $1 - \frac{1}{\delta} \geq 0$
   $\therefore \delta \geq 1, \alpha \geq 1$
   $\therefore \alpha \geq 1, \delta \geq 1, \beta \geq 0, \eta \geq 0$
2. α<0, $\gamma \leq 0$
3. $\alpha \geq 0, \beta$<0, η<0
4. $\alpha \geq 0, \beta \geq 0, \delta$<0